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Title

QUANTUM COMPUTING NOTES

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Contents

1	An Introduction to Quantum Computing	2
1.1	Debunking Quantum Computing Myths	2
1.2	Brief History of Quantum Computing	2
1.3	Current State of the Industry	3
1.3.1	Sector Investments	3
1.3.2	Technological Achievements	3
2	A review of complex linear algebra	5
2.1	Why Complex Numbers	5
2.2	Polar Coordinates and Exponential Form	6
2.3	Vectors and Matrices	7
2.4	Eigenvectors and Eigenvalues	8
3	Qubit	9
3.1	Probabilistic Algorithms	9
3.2	Classical vs. Quantum Randomness	9
3.3	The Quantum Bit (Qubit)	10
3.4	Qubit Measurement and Global Phase	10
3.5	Unobservability of Global Phase	10
3.6	Qubit Operations	11
3.7	Identity and Pauli Matrices	11
3.8	The Hadamard Transform	11
3.9	Quantum Evolution and Unitary Matrices	12
3.10	Picturing Qubits: The Bloch Sphere	12
4	Multiple Qubit	14
4.1	Basis Vectors and Classical Bit Systems	14
4.2	Representation of Multiple Classical Bits	14
4.3	Multiple Qubits and Tensor Products	14
4.4	Two-Qubit States	15
4.5	Quantum Entanglement	15
4.6	The EPR Paradox and Non-locality	16
4.7	The Controlled-NOT (CNOT) Gate	16
4.8	Linearity of CNOT and EPR Pair Generation	17
4.9	Quantum Circuit Notation and Multi-Qubit Circuits	17
4.10	Circuits for Entangled States	18
4.11	Tensor Product of Unitary Gates	18
	References	19

1 An Introduction to Quantum Computing

1.1 Debunking Quantum Computing Myths

- **Myth 1: Quantum computers will solve NP-complete problems efficiently.**
 - *Fact:* **False**, No efficient classical or quantum algorithm exists for NP-complete problems, and radical theoretical breakthroughs would be required to change this.
- **Myth 2: Quantum computers are too complicated to ever be built.**
 - *Fact:* **False**, Working Noisy Intermediate-Scale Quantum (NISQ) computers are operational today (e.g., systems built by IBM, Quantinuum, QuEra), including IBM's 1,121-qubit processor.
- **Myth 3: Quantum computing is just another fast hardware architecture (like GPUs or multi-core CPUs).**
 - *Fact:* **False**, It exploits quantum phenomena with no classical equivalent that cannot be simulated efficiently on classical machines, operating under a fundamentally different model of computation.
- **Myth 4: Quantum computing is hard (and requires knowing quantum physics).**
 - *Fact:* **Partially true**, but the mathematical foundation (formalized by John von Neumann) relies on complex linear algebra; the main current challenge is building proper high-level programming abstractions.
- **Myth 5: Quantum computers and classical computers solve the same problems.**
 - *Fact:* **True and False**, in terms of computability (they can simulate each other); however, classical simulation of quantum physics is exponentially slow, whereas quantum algorithms offer exponential speedups **for specific tasks**.

1.2 Brief History of Quantum Computing

- **1900:** Max Planck studies black-body radiation and discovers that electromagnetic radiation energy is emitted in discrete “quanta”.
- **1905:** Albert Einstein explains the photoelectric effect by proposing that light propagates as massless particles called photons.

- **1926:** Erwin Schrödinger formulates the time-dependent ($i\hbar \frac{d}{dt}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$) and time-independent ($\hat{H}|\Psi\rangle = E|\Psi\rangle$) equations describing the evolution of isolated quantum systems.
- **1927:** Werner Heisenberg introduces the uncertainty principle ($\sigma_x\sigma_p \geq \frac{\hbar}{2}$).
- **1932:** John von Neumann publishes *Mathematical Foundations of Quantum Mechanics*, establishing the theoretical foundation for quantum computing.
- **1963 / 1973:** Yves Lecerf and Charles Bennett introduce reversible Turing machines, proving that isolated quantum systems must undergo reversible computation.
- **1980 / 1982:** Paul Benioff, Yuri Manin, and Richard Feynman propose the first concepts of quantum computers to simulate physical quantum systems.
- **1994:** Peter Shor formulates the quantum factoring algorithm providing an exponential speedup over classical algorithms.
- **1996:** Lov Grover develops the quantum search algorithm providing a quadratic speedup over classical search.

1.3 Current State of the Industry

1.3.1 Sector Investments

- **Public Sector:** Substantial investments in quantum technologies, including the UK (~£1bn, highlighted in AUKUS), the EU (€1bn Quantum Flagship), the US, Germany/Bavaria (hundreds of millions of euros), and China (field leader in scientific journal publications).
- **Private Sector:** Commercial hardware developers selling systems (IBM, IonQ, IQM, Pasqal, QuEra, Quantinuum), private hardware initiatives (Google, Intel), research teams (Microsoft), cloud integrations (Amazon at Caltech), and financial applications in Quantum Machine Learning and simulation.

1.3.2 Technological Achievements

- **October 2019 (Google):** Demonstrated quantum supremacy using the 54-qubit *Sycamore* processor.
- **2023 (IBM):** Made 156-qubit systems publicly available and introduced the 1,121-qubit chip.
- **December 2023 (Harvard/MIT/QuEra):** Demonstrated error-corrected quantum algorithms on 48 logical qubits created from ~300 physical qubits

- **April 2024 (Microsoft & Quantinuum):** Created 4 highly reliable logical qubits from 30 physical qubits, achieving $\sim 99.8\%$ fidelity and an 800x reduction in error rate.
- **December 2024 (Google):** Achieved quantum error correction below the surface code threshold on the 105-qubit *Willow* system (executed random circuit sampling benchmark in under 5 minutes vs 10^{25} years required by supercomputers).
- **February 2025 (Microsoft):** Announced the *Majorana 1* topological chip utilizing Majorana particles for fault-tolerant quantum computing.
- **September 2025 (Harvard):** Neutral atom progress enabling the creation of 30,000 initialized qubits per second and coherent storage of 3,000 qubits for 2 hours.

2 A review of complex linear algebra

2.1 Why Complex Numbers

The set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ represents the most fundamental mathematical object, but it presents immediate limitations: a simple equation such as $x + 21 = 7$ has no solution in $\mathbb{N}$. By allowing negative numbers, the structure is extended to the set of integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$, where every equation of the form $x + a = b$ always has the integer solution $x = b - a$ (in our case $x = 7 - 21 = -14$).

Considering an equation like $22x + 7 = 0$, one notices that in general there are no integer solutions. It therefore becomes necessary to introduce the set of rational numbers $\mathbb{Q} = \{\dots, -\frac{2}{3}, -2, \dots, -\frac{1}{3}, -\frac{1}{2}, -1, 0, 1, \frac{1}{2}, \frac{1}{3}, \dots, 2, \frac{2}{3}, \dots\}$, in which the general equation $ax + b = 0$ (with $a \neq 0$) admits the rational solution $x = -b/a$ (in this specific case, $x = -7/22$).

Moving on to quadratic equations $ax^2 + bx + c = 0$ ($a \neq 0$ with a, b, c integers), the solutions are given by the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, valid provided that the discriminant $\Delta = b^2 - 4ac$ is non-negative ($\Delta \geq 0$). If Δ is the square of an integer, the solutions are rational; otherwise, irrational values may emerge (such as $\sqrt{2}$). When $\Delta < 0$, however, one obtains the square root of a negative number. For example, $\sqrt{-4} = \sqrt{4}\sqrt{-1} = 2\sqrt{-1} = 2i$, where $i = \sqrt{-1}$ defines the imaginary unit (such that $i^2 = -1$). A complex number is expressed in the form $a + ib$, with a and b real numbers (an example is $3.17 - 5i$).

When faced with higher-degree polynomials, such as $4x^5 + x^4 + 3.44x^2 + x - 7 = 0$, one might ask whether additional number sets are needed. The Fundamental Theorem of Algebra guarantees that complex numbers are sufficient: every equation of the form $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = 0$ (with $a_n \neq 0$ and all coefficients a_i complex) admits exactly n complex solutions.

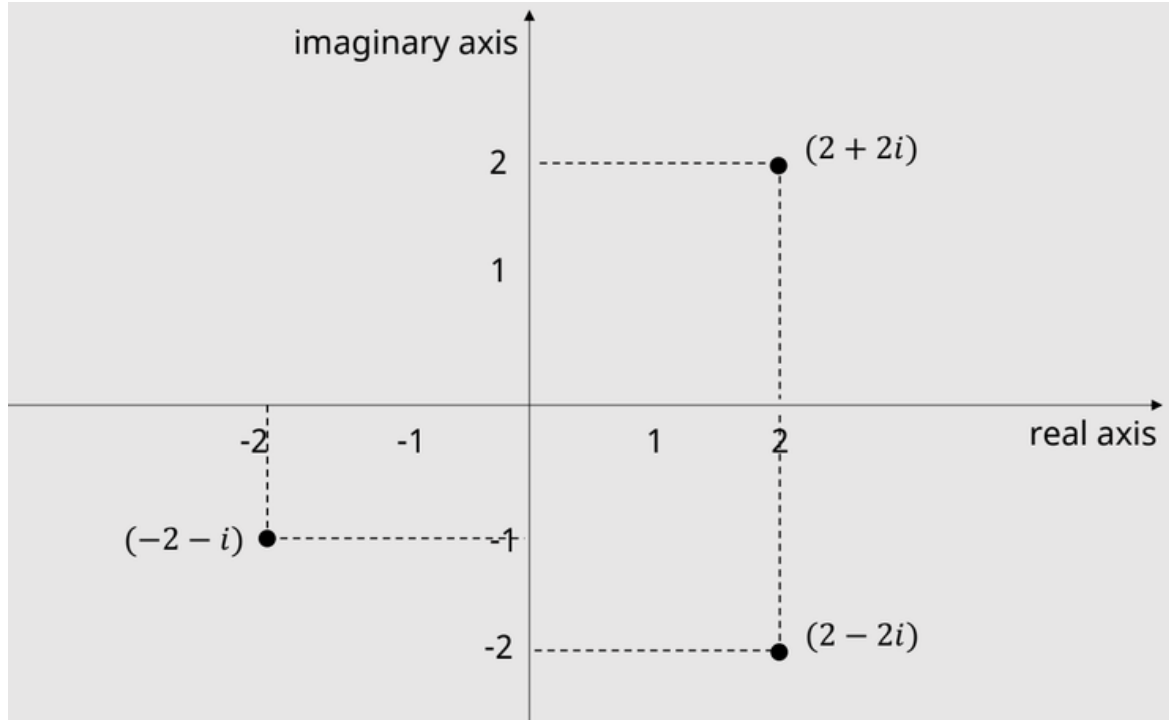
Complex Numbers and Basic Operations

The set of complex numbers is denoted by $\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$. Given a complex number $z = a + ib$, its real part is $\text{Re}(z) = a$, while its imaginary part is $\text{Im}(z) = b$. Note that the imaginary part of a complex number is a real number!

Complex numbers can be represented on the complex plane (or Argand-Gauss plane), where the horizontal axis measures the real part and the vertical axis measures the imaginary part. Points like $(2 + 2i)$, $(-2 - i)$, and $(2 - 2i)$ are placed precisely as orthogonal coordinate pairs on these axes.

Basic algebraic operations are defined as follows:

- **Addition:** $(a + ib) + (c + id) = (a + c) + i(b + d)$.
- **Multiplication:** $(a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc)$.
- **Conjugate:** The conjugate of $z = a + ib$ is $\bar{z} = a - ib$.



- **Modulus:** The modulus of $z = a + ib$ is the real number $|z| = \sqrt{a^2 + b^2}$, representing the distance of the element from the origin of the Cartesian plane. The relations $|zw| = |z||w|$ and $|z| = \sqrt{z\bar{z}}$ hold.
- **Division:** The division $\frac{z}{w}$ is calculated by multiplying both the numerator and the denominator by the conjugate of the denominator, $\bar{w}$.

2.2 Polar Coordinates and Exponential Form

A generic non-zero complex number $z = a + ib$ with modulus $|z| = \sqrt{a^2 + b^2}$ uniquely determines an angle $\theta \in (-\pi, \pi]$ such that $\cos \theta = \frac{a}{|z|}$ and $\sin \theta = \frac{b}{|z|}$. This allows writing z in polar coordinates as $z = |z|(\cos \theta + i \sin \theta)$. For example, $(2 + 2i) = |2 + 2i|(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}) = 2\sqrt{2}(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}})$.

Through the imaginary exponential defined by Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, one obtains the exponential form $z = |z|e^{i\theta}$. This notation simplifies calculations with powers:

- $z^2 = (|z|e^{i\theta})^2 = |z|^2 e^{i2\theta}$
- $\frac{1}{z} = z^{-1} = (|z|e^{i\theta})^{-1} = |z|^{-1} e^{-i\theta}$
- $\sqrt{z} = (|z|e^{i\theta})^{1/2} = \sqrt{|z|} e^{i\frac{\theta}{2}}$ (showing that the square root always exists in the complex field).

2.3 Vectors and Matrices

A complex vector is an ordered list of complex numbers. Given vectors $v = \begin{pmatrix} 4 \\ i \\ 2 + 3i \end{pmatrix}$ and $w = \begin{pmatrix} 0 \\ 1 \\ i \end{pmatrix}$, their sum is performed element by element, yielding $v + w = \begin{pmatrix} 4 \\ 1 + i \\ 2 + 4i \end{pmatrix}$.

Scalar multiplication also acts on each element: $v \cdot 2i = \begin{pmatrix} 8i \\ -2 \\ -6 + 4i \end{pmatrix}$.

The norm of a vector $v = (v_1, v_2, \dots, v_n)$ expresses its size or amplitude and is defined as $\|v\| = \sqrt{|v_1|^2 + |v_2|^2 + \dots + |v_n|^2} = \sqrt{\sum_{i=1}^n |v_i|^2}$ [cite: 1]. For example, the norm of $(4, i, 2 + 3i)$ is $\sqrt{|4|^2 + |i|^2 + |2 + 3i|^2} = \sqrt{16 + 1 + 13} = \sqrt{30}$.

Complex matrices are tables of complex numbers equipped with specific multiplication rules[cite: 1]. A matrix with m rows and n columns has dimension $m \times n$; when $m = n$, the matrix is called square. The 2×2 identity matrix is given by $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Addition between two matrices is allowed only if they have the same dimension and occurs by adding corresponding entries. For example, adding $\begin{bmatrix} 3 & i \\ 1 + i & 0 \\ -4 & 2 + 3i \end{bmatrix}$ and

$\begin{bmatrix} 1 & i \\ i & 10 \\ -4 & 2 - 3i \end{bmatrix}$ yields $\begin{bmatrix} 4 & 2i \\ 1 + 2i & 10 \\ -8 & 4 \end{bmatrix}$. Matrix addition enjoys the commutative property: $A + B = B + A$.

Row-by-column product between two matrices A (dimension $m \times n$) and B (dimension $p \times q$) requires $n = p$, producing a matrix of dimension $m \times q$. Multiplying a 3×2 matrix by a 2×2 matrix yields a 3×2 matrix; similarly, a 3×2 matrix multiplied by a 2×1 column vector returns a 3×1 vector[cite: 1]. Scalar multiplication simply multiplies every entry of the matrix by that scalar.

In general, matrix multiplication is NOT commutative ($A \cdot B \neq B \cdot A$). Considering matrices $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, the product $A \cdot B$ generates $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, while $B \cdot A$ produces $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ [cite: 1]. This non-commutative property is of fundamental importance in quantum physics and quantum computing.

2.4 Eigenvectors and Eigenvalues

A non-zero vector v is an eigenvector of a matrix A with eigenvalue λ if it satisfies $A \cdot v = \lambda \cdot v$. Equivalently, this corresponds to $(A - \lambda I) \cdot v = 0$, which admits non-zero solutions $v \neq 0$ if and only if the determinant is zero: $\det(A - \lambda I) = 0$.

As an example, the matrix $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ has eigenvalues $+1$ and -1 , with associated eigenvectors $\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ respectively. Indeed, for eigenvalue $+1$, verifying directly yields $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} -i^2 \\ i \end{bmatrix} = \begin{bmatrix} 1 \\ i \end{bmatrix}$.

In general, both eigenvectors v and eigenvalues λ can belong to the complex field, and it is possible for an eigenvalue to be associated with multiple eigenvectors (which form a linear subspace). In quantum mechanics, physical measurements are modeled by specific linear operators (represented by matrices) that always have real eigenvalues.

3 Qubit

3.1 Probabilistic Algorithms

Most standard computer algorithms are **deterministic**: for a given input, they will always produce the exact same output (assuming the program does not crash or hang). Note that multiple distinct input values (e.g., a, b, c) can map to the same output value, which is standard behavior (such as in a simple calculator program).

In contrast, **stochastic (probabilistic) algorithms** exhibit non-deterministic behavior. For a given input i , the program can produce different outputs with specific probabilities:

$$\text{input}_i \longrightarrow \begin{cases} \text{output}_{1,i} & \text{with probability } p_{1,i} \\ \text{output}_{2,i} & \text{with probability } p_{2,i} \\ \text{output}_{3,i} & \text{with probability } p_{3,i} \\ \dots & \\ \text{output}_{N,i} & \text{with probability } p_{N,i} \end{cases}$$

We assume that the number of possible outputs for any input is finite (N), although the set of potential inputs does not need to be finite. For any input i , the probabilities of all possible outputs must sum to 1:

$$\sum_{j=1}^N p_{j,i} = 1$$

This condition ensures probabilistic termination, meaning the program will output something with probability 1.

3.2 Classical vs. Quantum Randomness

In classical (non-quantum) computing, probabilistic behavior is introduced artificially using **Pseudo-Random Number Generators (PRNGs)**. PRNGs produce deterministic sequences of numbers designed such that, given one element, predicting the next is computationally difficult. However, because PRNGs rely on finite internal states, their generated sequences are inherently finite and eventually repeat, losing true randomness.

Quantum algorithms, on the other hand, are fundamentally probabilistic by the laws of physics. Quantum systems do not rely on PRNGs; their randomness is intrinsic to the nature of quantum measurements.

3.3 The Quantum Bit (Qubit)

A classical bit is binary and can take a value of either 0 or 1. The quantum equivalent basis states are represented by two-dimensional column vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The fundamental feature of a qubit is that its state can exist in a **superposition** of these basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{C}$ are complex numbers known as **probability amplitudes**. They satisfy the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

An important example of a qubit state in equal superposition is:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

3.4 Qubit Measurement and Global Phase

The "true" quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ cannot be observed directly. In principle, it is impossible to extract the exact values of the complex amplitudes α and β through a single measurement.

When a qubit is measured in the computational basis, the superposition collapses probabilistically:

$$\alpha|0\rangle + \beta|1\rangle \xrightarrow{\text{Measurement}} \begin{cases} |0\rangle & \text{with probability } |\alpha|^2 \\ |1\rangle & \text{with probability } |\beta|^2 \end{cases}$$

Immediately after measurement, the qubit state collapses completely into either $|0\rangle$ or $|1\rangle$.

3.5 Unobservability of Global Phase

For any complex number z with unit modulus ($|z| = 1$), and any complex number α , we have $|z\alpha| = |\alpha|$. Consequently, two qubit states that differ only by a overall complex phase factor z :

$$z\alpha|0\rangle + z\beta|1\rangle \quad \text{and} \quad \alpha|0\rangle + \beta|1\rangle$$

are physically identical and **cannot be distinguished** by any quantum measurement, as the output probabilities remain $|\alpha|^2$ and $|\beta|^2$.

3.6 Qubit Operations

All quantum operations (excluding measurement) are linear transformations. For a general transformation T :

$$T(\alpha|0\rangle + \beta|1\rangle) = \alpha T(|0\rangle) + \beta T(|1\rangle)$$

Every linear transformation on a two-dimensional complex vector space can be represented by a 2×2 matrix.

3.7 Identity and Pauli Matrices

The "do-nothing" transformation is represented by the 2×2 identity matrix:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I|0\rangle = |0\rangle, \quad I|1\rangle = |1\rangle$$

The quantum analogue of the classical NOT gate is the Pauli- X matrix (σ_x):

$$\text{NOT} = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Its action on basis vectors flips the state:

$$\sigma_x|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle, \quad \sigma_x|1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

Notice that $\sigma_x^2 = I$.

The other two Pauli matrices widely used in quantum physics and quantum computation are σ_y and σ_z :

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Their actions on the computational basis vectors are:

$$\begin{aligned} \sigma_y|0\rangle &= i|1\rangle, & \sigma_y|1\rangle &= -i|0\rangle \\ \sigma_z|0\rangle &= |0\rangle, & \sigma_z|1\rangle &= -|1\rangle \end{aligned}$$

All Pauli matrices satisfy $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I$.

3.8 The Hadamard Transform

The Hadamard gate H is one of the most critical transformations in quantum computing:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Applying H to the basis state $|0\rangle$ creates an equal superposition:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Similarly, applying H to $|1\rangle$ gives:

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

If we apply a measurement to the state $H|0\rangle$, we obtain $|0\rangle$ with probability $1/2$ and $|1\rangle$ with probability $1/2$. Thus, the Hadamard transform applied to a classical basis state acts as a fundamental, truly random bit generator.

3.9 Quantum Evolution and Unitary Matrices

Not every matrix represents a valid quantum transformation. Allowed physical transformations must preserve the norm of the input vector.

Given $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, its norm is $\| |\psi\rangle \| = \sqrt{|\alpha|^2 + |\beta|^2} = 1$. Since the total probability of all measurement outcomes must equal 1, quantum transformations must maintain this normalization.

Matrices that preserve vector norms are called **unitary matrices**.

Definition 3.1. A matrix U is unitary if and only if $\|Uv\| = \|v\|$ for any state vector v (or equivalently, $U^\dagger U = I$, where $U^\dagger$ is the conjugate transpose of U).

Unitary transformations preserve overall probabilities. It can be easily verified that the Identity matrix, all Pauli matrices $(\sigma_x, \sigma_y, \sigma_z)$, and the Hadamard matrix H are all unitary.

3.10 Picturing Qubits: The Bloch Sphere

Any normalized single-qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ (with $|\alpha|^2 + |\beta|^2 = 1$) can be parameterised using polar coordinates.

Expressing α and β in polar form, the state can be rewritten as:

$$|\psi\rangle = e^{i\gamma} \left(\cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle \right)$$

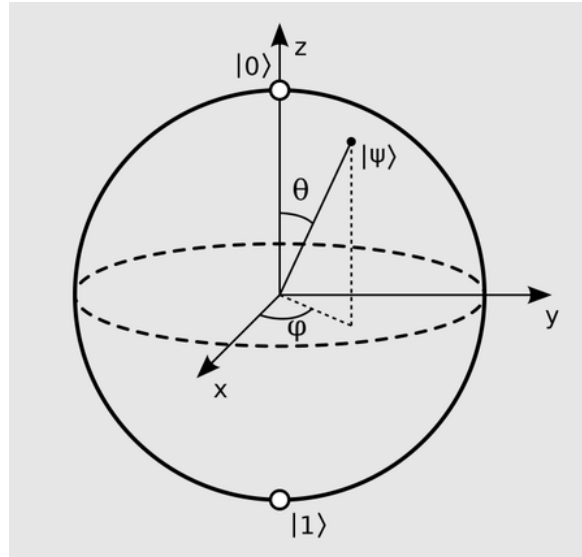
where γ, θ, φ are real parameters.

Since the global phase factor $e^{i\gamma}$ has modulus $|e^{i\gamma}| = 1$, it produces no observable physical effects and can be ignored. This simplifies the canonical representation of a single qubit to:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

where $0 \leq \theta \leq \pi$ and $0 \leq \varphi < 2\pi$.

Geometrically, this parameterization maps state vectors onto the surface of a three-dimensional unit sphere known as the **Bloch Sphere**:



- The North Pole ($\theta = 0$) corresponds to state $|0\rangle$.
- The South Pole ($\theta = \pi$) corresponds to state $|1\rangle$.
- The angle θ represents the polar angle along the z -axis, while φ represents the azimuthal angle in the x - y plane.

Important Note: The Bloch sphere provides an intuitive visual model exclusively for single-qubit states. There is no simple equivalent representation for systems composed of two or more qubits due to quantum entanglement and the exponential growth of the underlying vector space.

4 Multiple Qubit

4.1 Basis Vectors and Classical Bit Systems

Any vector in a vector space can be decomposed as a linear combination of basis vectors. For example:

$$\begin{pmatrix} 3 \\ -2 \\ 1+i \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + (-2) \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + (1+i) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

This property holds for complex vector spaces of arbitrary dimensions (including infinite dimensions with appropriate conditions).

In general, an n -dimensional vector can be expressed as:

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \alpha_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \cdots + \alpha_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} = \sum_{i=1}^n \alpha_i e_i$$

where e_i is the basis vector with 1 in the i -th entry and 0 elsewhere. These diagonal basis vectors typically correspond to classical states.

4.2 Representation of Multiple Classical Bits

To represent n classical bits, one lists them in a vector or array of n elements. Formally, given the set $\mathcal{B} = \{0, 1\}$, the state space is defined via the Cartesian product $\mathcal{B}^n = \mathcal{B} \times \cdots \times \mathcal{B}$. A classical state is an n -tuple $s = (b_1, b_2, \dots, b_n) \in \mathcal{B}^n$.

Key characteristics of classical bit representations:

- The length of the vector grows linearly with n .
- Vector components are mutually independent (changing one component does not affect the others).

4.3 Multiple Qubits and Tensor Products

To describe n quantum bits (qubits), the tensor product ($\otimes$) is required.

Single-qubit basis states are defined as $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. A general single-qubit state is a superposition $\alpha_0|0\rangle + \alpha_1|1\rangle = \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix}$ satisfying $|\alpha_0|^2 + |\alpha_1|^2 = 1$.

4.4 Two-Qubit States

The tensor product of two single-qubit states is computed as:

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} \otimes \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} \alpha_0\beta_0 \\ \alpha_0\beta_1 \\ \alpha_1\beta_0 \\ \alpha_1\beta_1 \end{pmatrix} = \alpha_0\beta_0|00\rangle + \alpha_0\beta_1|01\rangle + \alpha_1\beta_0|10\rangle + \alpha_1\beta_1|11\rangle$$

where $|00\rangle = (1, 0, 0, 0)^T$, $|01\rangle = (0, 1, 0, 0)^T$, $|10\rangle = (0, 0, 1, 0)^T$, and $|11\rangle = (0, 0, 0, 1)^T$.

Properties and examples of tensor products for qubits:

- The tensor product extends naturally to any number n of qubits.
- The dimension of the resulting state vector grows exponentially (2^n) with the number of qubits.
- Tensor products yield valid quantum states satisfying normalization: $\sum_{i,j \in \mathcal{B}} |\alpha_i\beta_j|^2 = 1$.

- Example 1: $|0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle$.

- Example 2: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

The state $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ represents an equal superposition over all 2-bit classical states. Although it appears complex, it can be factored into two independent single-qubit states.

4.5 Quantum Entanglement

According to quantum mechanics, any normalized linear combination of the form:

$$\alpha_0|00\rangle + \alpha_1|01\rangle + \alpha_2|10\rangle + \alpha_3|11\rangle \quad \text{with} \quad \sum_{i=0}^3 |\alpha_i|^2 = 1$$

is a valid state for a 2-qubit system.

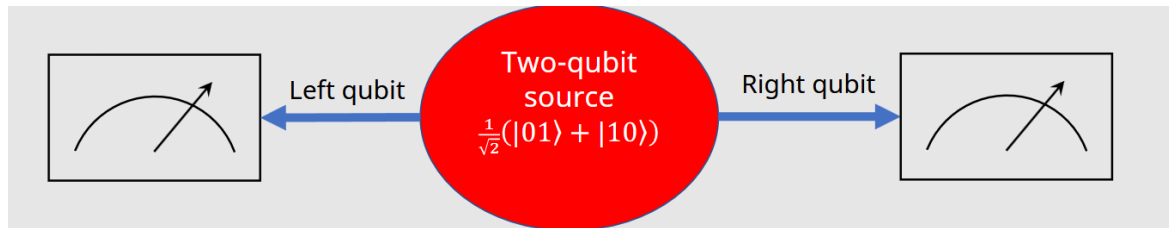
Consider the state $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$:

- It cannot be decomposed into two independent local qubit states; it describes the system as an indivisible whole (an *entangled state*).

- It forms an EPR pair (Einstein, Podolsky, Rosen, 1935).
- Measurement of either qubit collapses the system into state $|01\rangle$ with probability $1/2$ or state $|10\rangle$ with probability $1/2$.
- Measuring one qubit immediately reveals the state of the other qubit (which is not the case for separable states like $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$).

4.6 The EPR Paradox and Non-locality

If two entangled qubits fly apart, they maintain their anti-correlation regardless of distance:

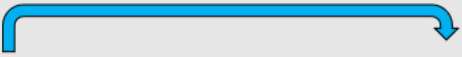


- Einstein referred to this non-local behavior as “spooky action at a distance”.
- Entanglement has been experimentally validated over distances exceeding 1,000 km.
- The 2022 Nobel Prize in Physics was awarded to Alain Aspect, John Clauser, and Anton Zeilinger for experiments verifying quantum entanglement.

4.7 The Controlled-NOT (CNOT) Gate

The Controlled-NOT (CNOT) gate operates on two qubits: a control qubit a and a target qubit b .

- Classical functionality: `if a then b := NOT(b) else skip.`

			
Target bit	a	b	$CNOT(a, b)$
	0	0	0
	0	1	1
Control bit	1	0	1
	1	1	0

- Classical truth table: $(0,0) \rightarrow (0,0)$, $(0,1) \rightarrow (0,1)$, $(1,0) \rightarrow (1,1)$, $(1,1) \rightarrow (1,0)$.
- Quantum operation: $CNOT(|a,b\rangle) = |a, a \oplus b\rangle$, where $\oplus$ denotes XOR.
- Action on basis states:

$$\begin{aligned} CNOT(|00\rangle) &= |00\rangle, & CNOT(|01\rangle) &= |01\rangle, \\ CNOT(|10\rangle) &= |11\rangle, & CNOT(|11\rangle) &= |10\rangle. \end{aligned}$$

- 4×4 Matrix representation:

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

- Example matrix-vector calculation for $|10\rangle$:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |11\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

4.8 Linearity of CNOT and EPR Pair Generation

Since CNOT is a linear operator:

- $CNOT\left(\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)\right) = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$.
- $CNOT\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle\right) = CNOT\left(\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)\right) = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ (creates an EPR pair!).

4.9 Quantum Circuit Notation and Multi-Qubit Circuits

Key quantum gates used in circuit diagrams:

- **NOT gate (X):** Maps $|a\rangle \rightarrow |\neg a\rangle$.
- **Hadamard gate (H):** Maps $|a\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + (-1)^a|1\rangle)$.
- **CNOT gate:** Maps $|a,b\rangle \rightarrow |a, a \oplus b\rangle$.
- **Measurement gate:** Measures $\alpha_0|0\rangle + \alpha_1|1\rangle$, yielding $|0\rangle$ with probability $|\alpha_0|^2$ or $|1\rangle$ with probability $|\alpha_1|^2$.

4.10 Circuits for Entangled States

1. **Creating an EPR Pair:** Starting with $|0\rangle$ (control) and $|1\rangle$ (target), apply an H gate to the control qubit yielding $\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$, followed by a CNOT gate to produce the EPR state $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$.
2. **Creating the GHZ State:** The Greenberger–Horne–Zeilinger (GHZ) state is a 3-qubit entangled state. Starting with input $|000\rangle$, apply an H gate to qubit 1, a CNOT gate between qubit 1 (control) and qubit 2 (target), and a CNOT gate between qubit 2 (control) and qubit 3 (target). The resulting state is:

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

4.11 Tensor Product of Unitary Gates

For two 2×2 matrices A and B , their tensor product is a 4×4 matrix given by the Kronecker product:

$$A \otimes B = \begin{pmatrix} a_{1,1}B & a_{1,2}B \\ a_{2,1}B & a_{2,2}B \end{pmatrix}$$

Applying $H \otimes H$ to $|00\rangle$ produces the superposition state $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

Given $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, the explicit matrix product $H \otimes H$ is calculated as:

$$H \otimes H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1H & 1H \\ 1H & -1H \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

References

- [1] Leslie Lamport. *LaTeX: a Document Preparation System*. Addison Wesley, Massachusetts, 2 edition, 1994.